

Vineet Kumar Upadhyay

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Backend Engineer | Java & Spring Boot | PostgreSQL | Distributed Systems

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Professional Summary

Final-year B.Tech CSE student building CYPR, a cybersecurity SaaS platform, with hands-on experience designing secure REST APIs, real-time WebSocket systems, and JWT/OAuth authentication infrastructure in Java and Spring Boot, plus the vanilla JavaScript frontends that consume them. Strong foundation in data structures, algorithms, and system design.

Technical Skills

- **Languages & Core CS:** Java, JavaScript, SQL, Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
- **Frameworks:** Spring Boot, Spring Security, Hibernate/JPA
- **Frontend:** HTML, CSS, JavaScript (vanilla), REST API integration, responsive UI
- **Databases & Caching:** PostgreSQL, MySQL, Redis, Caffeine Cache, HikariCP Tuning
- **Cloud & DevOps:** AWS (EC2, RDS, ALB), Docker, Kubernetes, Git, CI/CD, Vercel
- **Testing:** JUnit, Mockito, Integration Testing
- **Backend Concepts:** REST APIs, WebSockets (STOMP), JWT/OAuth 2.0, Multithreading, RBAC, Rate Limiting, Microservices

Projects

CYPR — Enterprise Cybersecurity Suite Platform

Tech Stack: *Java, Spring Boot, PostgreSQL, JavaScript, WebSockets, JWT/OAuth 2.0, Docker*

- Built **VAJRA**, a self-contained phishing detection engine combining 15+ independent heuristic signals (Shannon entropy, N-gram modeling, Levenshtein distance, homoglyph and combosquatting detection) with zero external API dependency.
- Designed and deployed the product frontend (HTML, CSS, vanilla JavaScript) on Vercel, wiring dashboard, admin, and auth views directly to the Spring Boot REST API layer.
- Implemented JWT-based auth and OAuth 2.0 social login with step-up MFA and dual-counter (session + user-level) logout; hardened flows against account-takeover and credential-stuffing.
- Eliminated IDOR vulnerabilities across the API by deriving user identity from JWT claims instead of client-supplied path parameters.
- Designed a CI/CD Orchestrator with bounded thread pools for async build execution, real-time WebSocket log streaming, and sanitized log output to prevent injection.
- Enforced RBAC on admin endpoints, idempotency keys with request-hash conflict detection, and optimistic locking on high-contention entities to prevent race conditions.
- Tuned HikariCP connection pooling on PostgreSQL/AWS RDS and used Caffeine Cache for request rate limiting under concurrent load.

FocusHub — Browser Productivity Suite (Chrome Extension)

Tech Stack: *JavaScript, Manifest V3, Web Audio API, Chrome Storage API*

- Built a Manifest V3 Chrome extension in vanilla JavaScript for site blocking and focus tracking, using the Web Audio API for an in-browser soundscape mixer.
- Designed the extension's popup and options UI end-to-end (HTML/CSS/JS) with persistent state via the Chrome Storage API.

Education

IEC College of Engineering and Technology	Greater Noida, UP
<i>B.Tech in Computer Science and Engineering</i>	2023 – 2027
Intermediate (Class XII)	UP Board
<i>Senior Secondary Education</i>	2020 – 2022
High School (Class X)	UP Board
<i>Secondary Education</i>	2019 – 2020

Achievements

- **National Finalist, ABU Robocon India (2025):** Competed among top robotics teams nationally in autonomous/manual robot design and control.
- **Tata Imagination Challenge (2025):** Participated in the Tata Quiz, organized by Tata Group via Unstoppable.

Certifications

- **Java Programming Training:** IEC College of Engineering & Technology, in association with Technoledge Eduresearch Pvt. Ltd. (Apr – Jun 2025)
- **Postman API Fundamentals (Student Expert):** LetsUpgrade x NSDC x GDG MAD (Nov 2024)